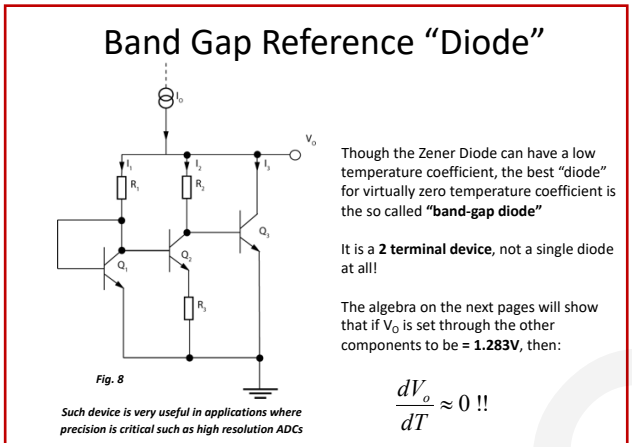
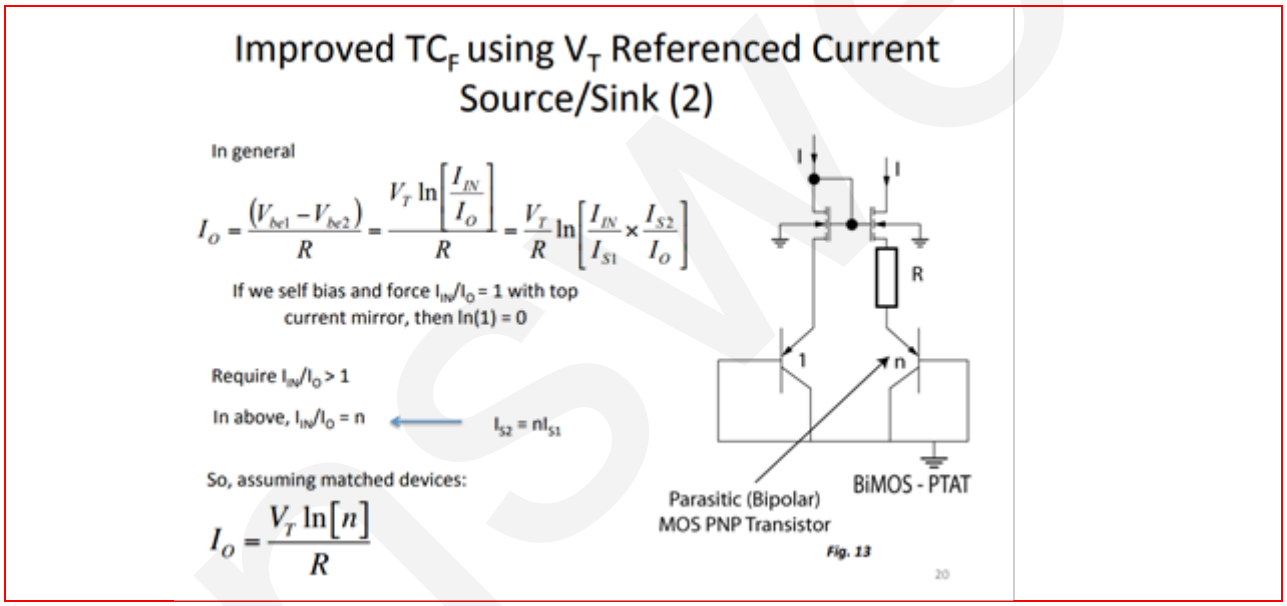
1. (a) Figures 1(a) and 1(b) show two popular biasing schemes typically used in analogue integrated systems. Briefly outline the main feature of each of these circuits.

[4]





For the bandgap voltage reference circuit of Figure 1(a), show that $\delta V_0 / \delta T = 0$ (where T is temperature) if $(R_2/R_3) ln$ [I_1/I_2] = 29 for V_0 = 1.283 V. Assume the temperature coefficient of V_{BE} to be -2.5mV/°C, the collector current of transistor Q_3 is 100µA and the device saturation current is I_S = 1.2 x$10^{-13}A$. Boltzmann's constant k = 1.38 x$10^{-23}J/K$ and the electron charge is q = 1.6 x$10^{-19}C$.

[7]

Using the equation $I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)$ and given that all the 3 BJTs are well matched and at the same temperature T, then from the circuit:

$V_{BE1} = V_{BE2} + I_2 R_3$ **Note: all β >> 1**

Then:

$$V_{BE1} - V_{BE2} = V_T \ln\left(\frac{I_1}{I_2}\right) = I_2 R_3$$

So:

$$V_O = V_{BE3} + \left(\frac{R_2}{R_3}\right) V_T \ln\left(\frac{I_1}{I_2}\right)$$

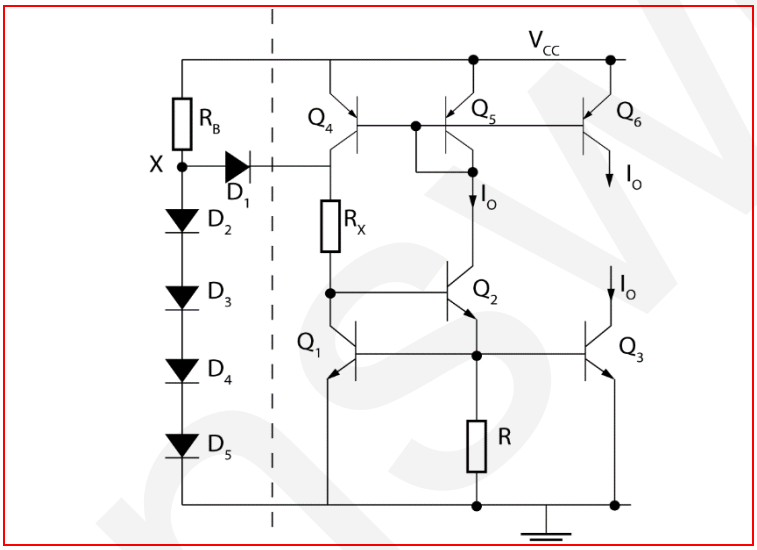
$$\frac{dV_O}{dT} = \frac{dV_{BE3}}{dT} + \frac{R_2}{R_3} V_T \frac{d}{dT} \ln\left(\frac{I_1}{I_2}\right)$$

13

(b) Figure 1(c) below shows a single stage of a differential amplifier.

(i) Draw a circuit schematic for the current source I_0 that ensures the differential gain is independent of temperature and power supply variation. Include any start up circuits that may be required.

[4]



(ii) For your design, show that the gain bandwidth product is independent of temperature.

[3]

$$GBW = \frac{g_{m2}}{2\pi C_L}$$

$$g_m = \frac{I_0}{V_t}$$

$$I_0 = \left(V_{be1} - V_{be2}\right) \Big/ R = V_T \ln\left[I_{in} \Big/ I_0\right] \Big/ R$$

$$I_{in} \Big/ I_0 = 2$$

$$g_m = \frac{V_T \ln[2]}{2\pi V_T} = \frac{\ln[2]}{2\pi R}$$

$$GBW = \frac{\ln[2]}{2\pi R C_L}$$

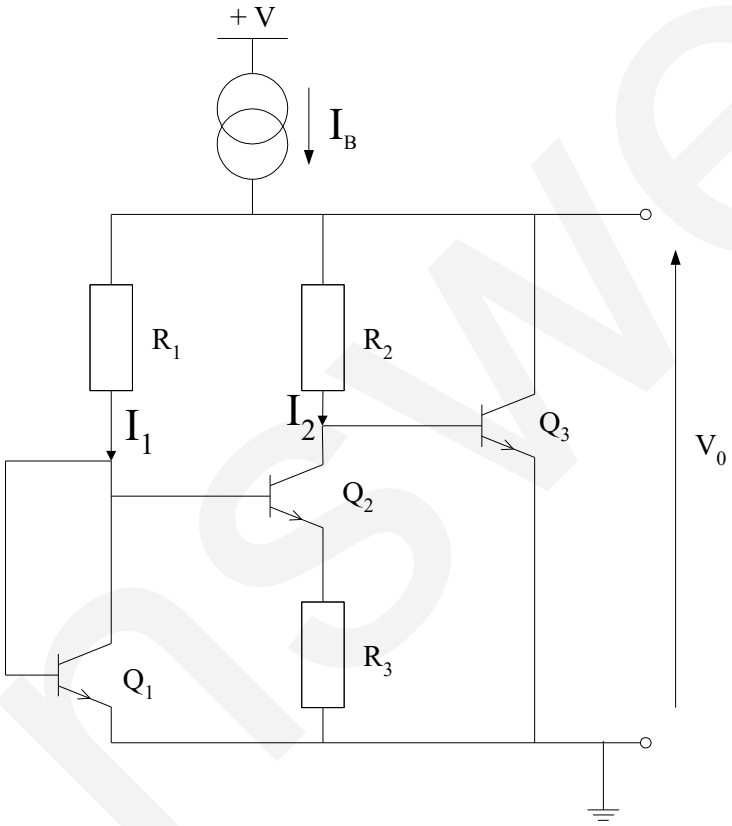


Figure 1(a)

Answers

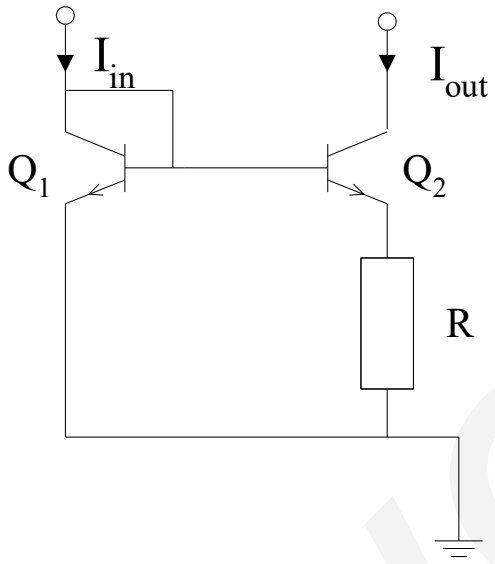


Figure 1(b)

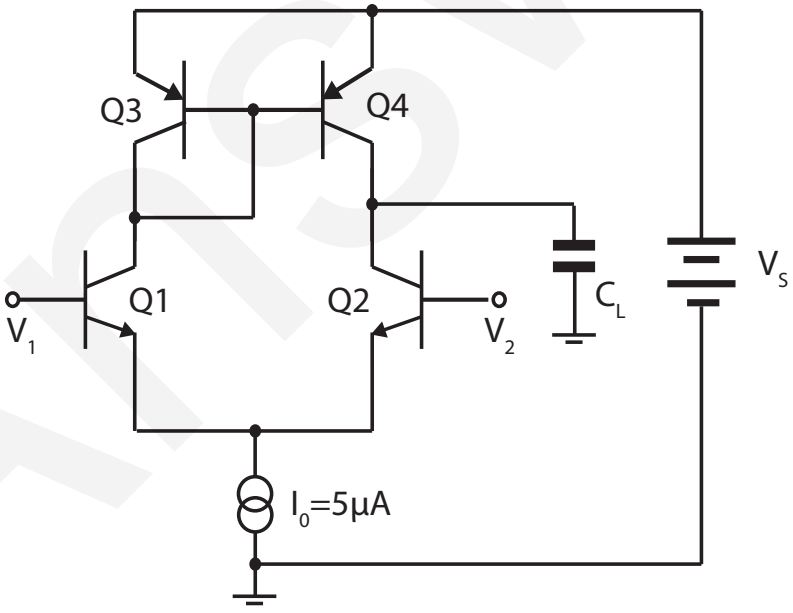
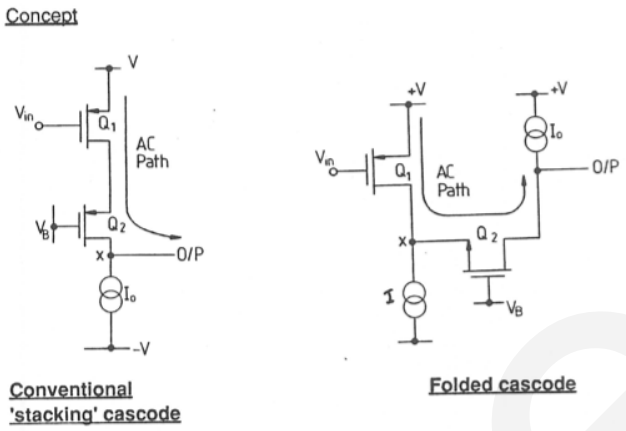


Figure 1(c)

Answers

2. (a) Figure 2(a) shows a folded cascode connection. What is the main advantage of this design over the more classical cascode connection? Show via a brief sketch how the architecture of Figure 2(a) can be used to form the basis of a single stage fully differential folded cascode operational amplifier (op-amp) with common-mode feedback included.

[6]



In the folded cascode we are unstacking the conventional cascode and spreading it out. The AC current path is folded and this allow a reduction in power supply. The conventional cascode requires a 2-stage architecture and since the impedance at (x) is high requires internal compensation. The folded cascode can be used as a single stage architecture, node x is low impedance so the only high impedance node will be at the output.
Gain of the folded cascode Av=gm1/Go.

[4]

(b) Figure 2(b) shows a two-stage CMOS op-amp. Estimate the low-frequency differential voltage gain and gain-bandwidth product of the amplifier. Aspect ratios of all devices are shown on the circuit. Assume all bulk effects are negligible. Device model parameters are given below. What is the main advantage and disadvantage of a single-stage compared to a two-stage op-amp ?

[8]

Answers

$$Av1 = -gm2/(go2+go4)$$
$$(go2+go4) = I_{D2}(\lambda_N + \lambda_P) = 5\times10^{-5}\times0.05 = 2.5\times10^{-7}\Omega^{-1}$$
$$gm2 = 2\sqrt{\beta_2 I_D}$$
$$\beta_2 = \frac{K_N}{2}\left(\frac{W}{L}\right)_2 = 7.5\times10^{-5}A/V$$
$$gm2 = 3.87\times10^{-5}S$$
$$Av1 = -154.9$$
$$Av2 = -gm6/(go7+go6)$$
$$(go7+go6) = I_{D6}(\lambda_N + \lambda_P) = 20\times10^{-5}\times0.05 = 10\times10^{-7}\Omega^{-1}$$
$$gm6 = 2\sqrt{\beta_6 I_{D6}}$$
$$\beta_6 = \frac{K_N}{2}\left(\frac{W}{L}\right)_6 = 1.6\times10^{-4}A/V$$
$$gm6 = 1.13\times10^{-4}S$$
$$Av2 = -113$$
$$A_{total} = Av1\times Av2 = 17503$$
$$G.Bp = gm2/2\pi C_c = 12.32MHz$$

(c) Explain why a resistor in series with the compensation capacitor *C* in Figure 6(b) can significantly improve the amplifier's phase margin.

[3]

Nulling Resistor (2)

Lets do better and instead of cancel Z use it to our advantage!

How?

We can attempt to **cancel P_2** simply by adjusting R_z until $Z = P_2$

i.e. mathematically the zero is moved in to the left-hand plane and so

$$\frac{V_{O2}}{V_{in}} = \left(\frac{A_1A_2}{1+\frac{jf}{f_{p1}}}\right)\left(\frac{1+\frac{jf}{f_Z}}{1+\frac{jf}{f_{p2}}}\right)$$

Achieving this, we would be left with a single dominant pole at f_{p1} and excellent phase margin!

(d) Draw an equivalent circuit for the PSRR of the op-amp and explain why it is low.

[3]

Answers

Power Supply Rejection Ratio in Simple MOS Op-Amps (3)

Reduces to ≈

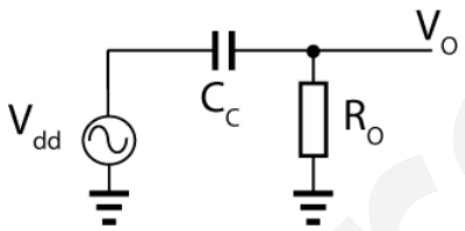


Fig. 18

V_{dd} coupling from supply via Q_6 and C_C to output.

i.e. since I_7 constant **V_{GS6} looks like battery**.

To improve PSRR reduce or eliminate C_C.

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CMOS TRANSISTOR PARAMETERS

MODEL PARAMETERS	*Kp* ($\mu A/V^2$)	λ (V^{-1})	V_{T0} (V)
PMOS	20	0.03	-0.8
NMOS	30	0.02	1.0

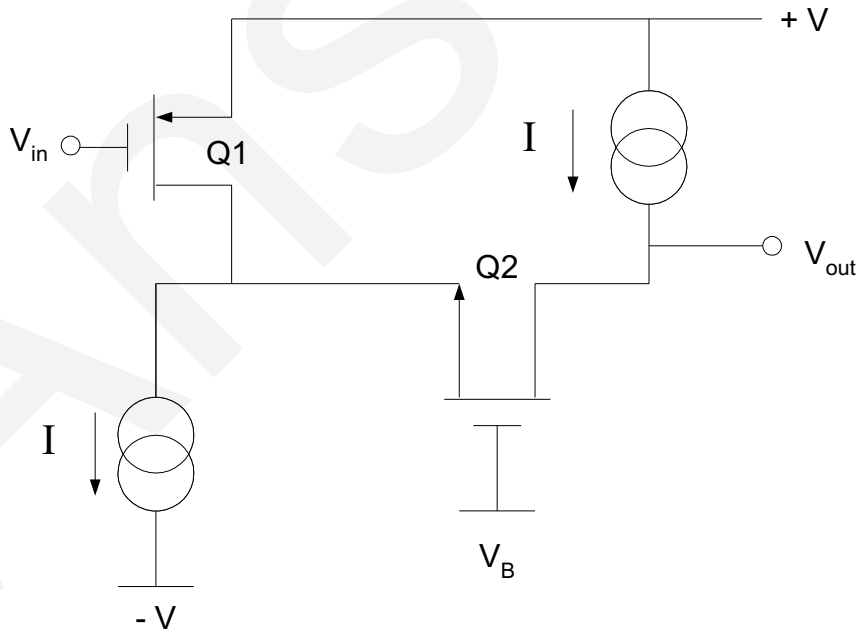


Figure 2(a)

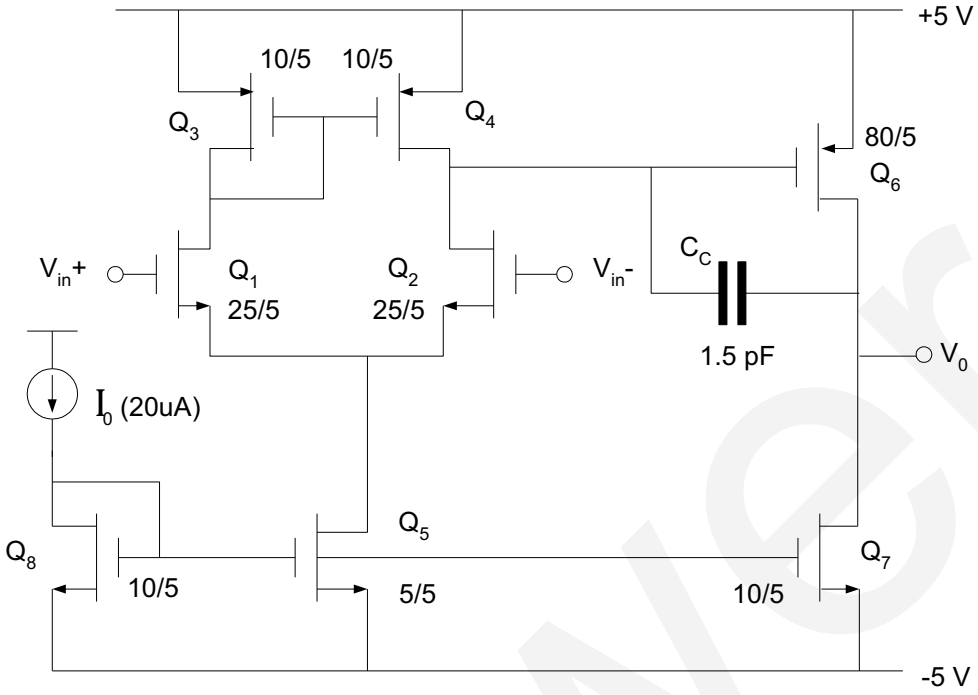


Figure 2(b)

Answers

3. (a) Sketch circuits for three integrators – A fully differential continuous-time integrator, a switched-capacitor integrator and a switched-current integrator.

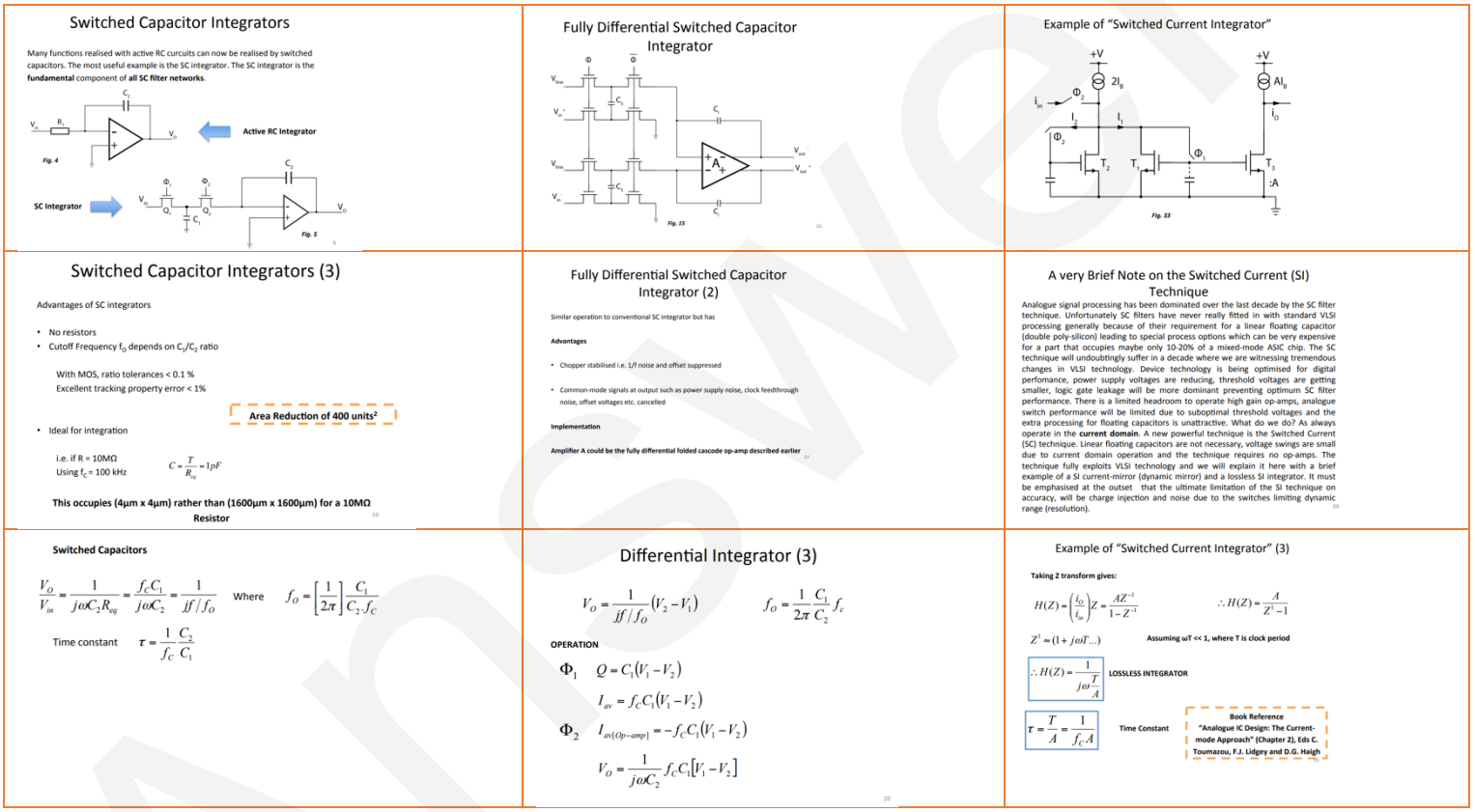
[4]

(b) Explain the advantages and disadvantages of each of the integrators.

[6]

(c) Derive the integrator transfer function of each of the integrators. For both the switched-capacitor and switched-current integrators, what limits the maximum operating frequency?

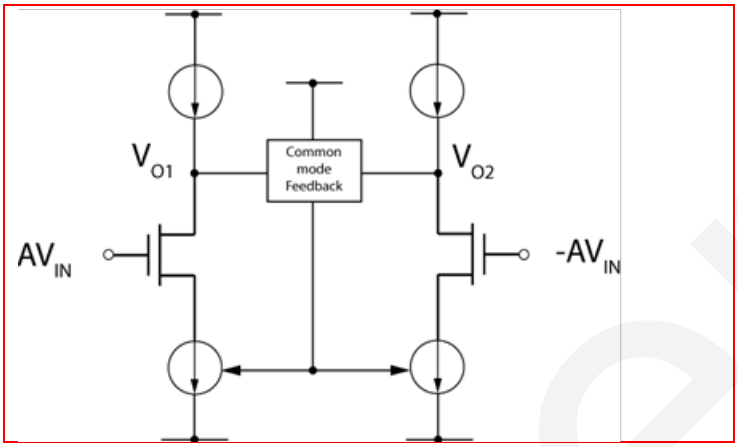
[10]



4. Figure 4 shows a differential opamp which is based upon a single-stage fully differential design.

(a). Sketch the circuit diagram of a single-stage fully differential op-amp including common mode feedback.

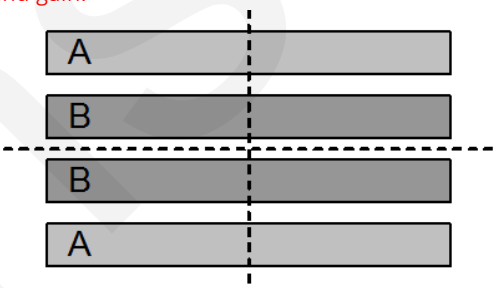
[6]



(b). Explain the technique used for efficient layout when designing transistor-based amplifiers.

[4]

Mismatches like λ (the Vds difference), threshold voltage VT and W/L may cause errors in offset and gain. A proper layout may help reduce the effect of such mismatches as well as gradients. Common centroid design splits the transistors over a common centroid or axis of symmetry. For example, the elements A and B in an array may be inter-digitated as in figure below. The input differential pair transistors in the Op-amp are the critical ones for the layout as they affect the offset and gain.



(c). (i) Transistors Q1 and Q2 of Figure 4 could be considered to form a linear differential resistor. Derive the equation for linear differential resistance. How does the body effect affect the linearity of each of these transistors?

[5]

Answers

$$I_1 = 2\beta[(V_g - V_x - V_T)(V_{in}/2 - V_x) - 1/2(V_{in}/2 - V_x)^2]$$
$$I_2 = 2\beta[(V_g - V_x - V_T)(-V_{in}/2 - V_x) - 1/2(-V_{in}/2 - V_x)^2]$$
$$R = 2V_{in}/(I_1 - I_2)$$
$$R = \frac{1}{\beta(V_g - V_T)}$$

(ii) Assuming the time constant of the circuit is τ=10ms, calculate the value of feedback capacitor, C, assuming input transistors Q1 and Q2 are operating in the triode region and have W/L=5, K=20uA/V2, Vg=1V, Vt=0.5V and node x is at a common mode of 0 V. You may neglect any body effects.

[5]

Time constant τ=RC=10ms
R=1/20ux5(1-0.5)/2=40 KΩ
C=τ/R=10m/40K=2.5x10^{-7} f

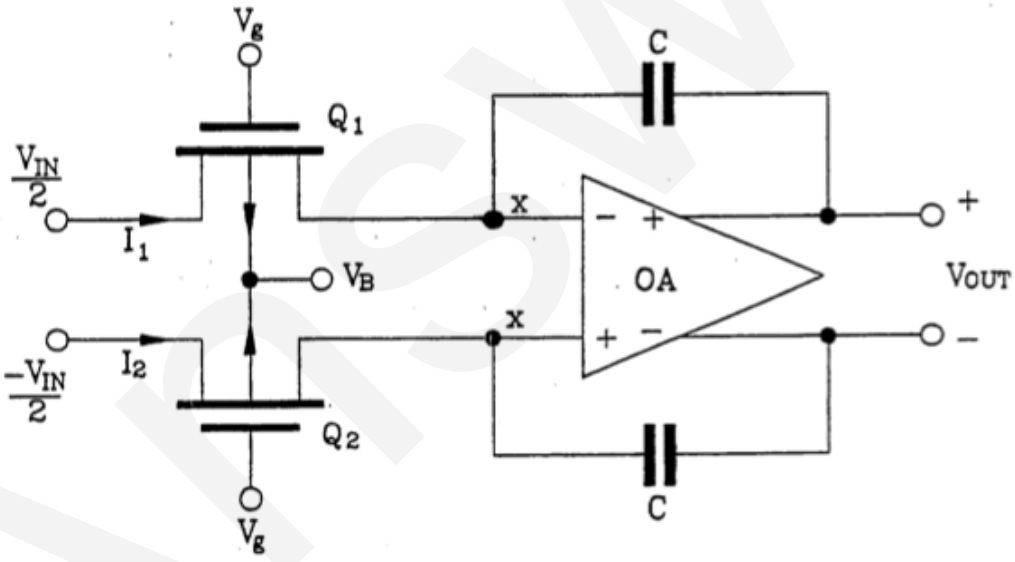
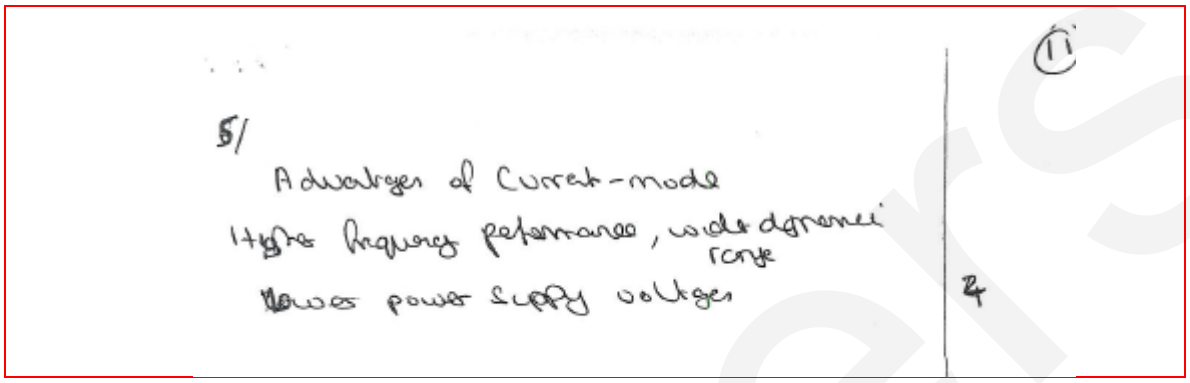


Figure 4.

5. (a) Give two advantages of current-mode analogue signal processing compared to traditional voltage-mode processing.

[2]



(b) With the aid of a suitable macromodel, explain the theoretical concept of current feedback and how it results in constant bandwidth amplification. Using a current-feedback op-amp, design a closed-loop non-inverting gain stage with a bandwidth of 10 MHz for a fixed voltage gain of 100. Assume an internal compensation capacitance of 4pF and that the open-loop transresistance gain of the amplifier is very much larger than the amplifier feedback resistor.

[6]

Answers

(ii)

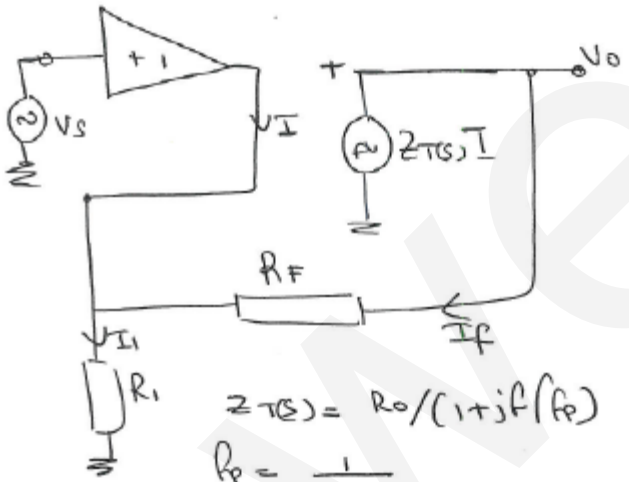
5/

Advantages of Current-mode

Higher frequency performance, wider dynamic range

Lower power supply voltages

4



$Z_{TIS} = R_0/(1+jf(f_p))$

$f_p = \frac{1}{2\pi R_0 C}$

C = compensation capacitor

3 equations

$I_f = (V_o - V_s)R_F$ — (1)

$I_1 = V_s/R_1$ — (2)

$V_o = Z_{TIS} I = Z_{TIS}[I_1 - I_f]$ — (3)

8

(4)

(4)

(12)

subs ① and ② into ③ gives,

$$(V_o/V_s) = (1+R_F/R_1)\,Z_{(s)}/(R_F+Z_{(s)})$$

subs for $Z_{(s)}$

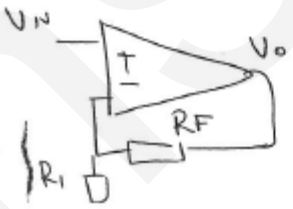
$$(V_o/V_s)_{sw} = \underbrace{(1+R_F/R_1)\left[\frac{R_o}{R_o+R_F}\right]}_{\text{gain}} \times \frac{1}{\left(1+jf/f_p\underbrace{\left[\frac{R_o+R_F}{R_F}\right]}_{BW}\right)}$$

Assuming $R_o \gg R_F$ then

Closed loop gain $= (1+R_F/R_1)$ — *

Closed loop Bandwidth $= \frac{f_p R_o}{R_F} = \frac{1}{2\pi R_F C}$ ✓✓.

Here R_F sets the amplifiers constant BW and R_1 chosen to set the gain



$BW = \frac{1}{2\pi R_F C} = 10\text{MHz}$

$\therefore$ given $C = 4\text{pF}$

$R_F = 3.98\text{k}\Omega$ B

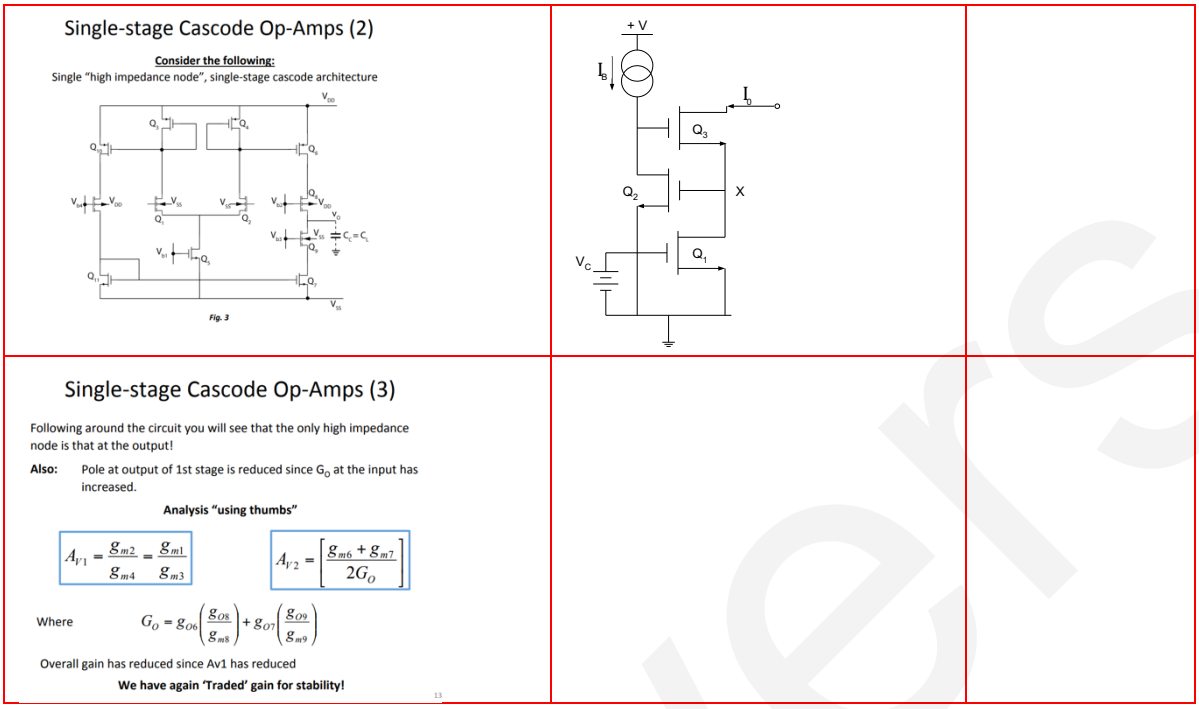
Since $A = (1+R_F/R_1) = 100$

$R_1 = 43.5\Omega$

(c) In order to improve the voltage gain of single stage amplifiers the cascading technique was introduced.

Please sketch suitable circuits for a single cascode amplifier, a regulated cascode amplifier and a high swing cascode amplifier and derive the output resistance of each.

[6]



(d) Sketch a typical architecture of a current-mode algorithmic analogue to digital converter and explain its principles of operation.

[6]

Answers

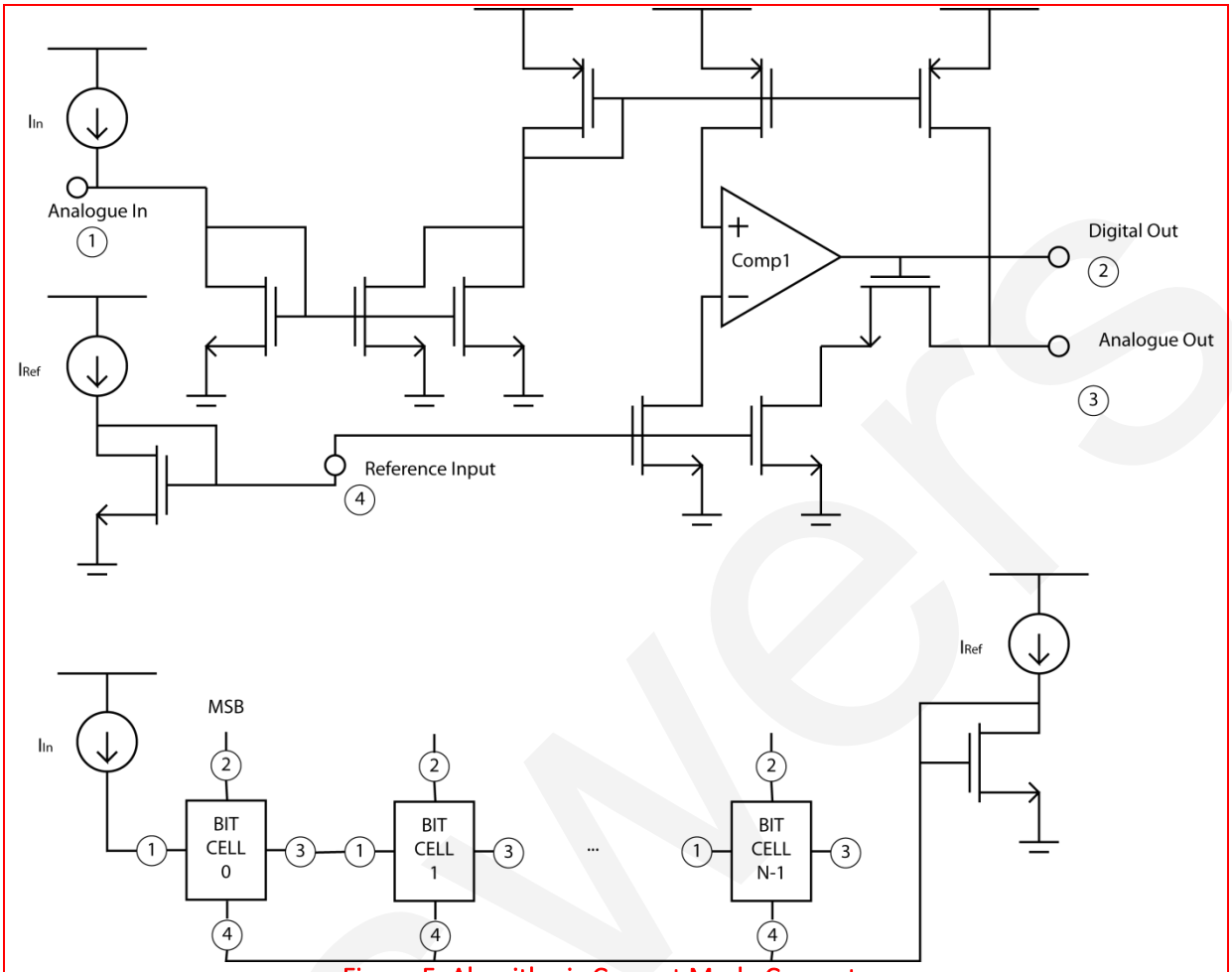


Figure 5. Algorithmic Current Mode Converter.

If 2Iin<IRef, comp goes low, digital output=0;
If 2Iin>IRef, comp output goes high, digital output=1
Analogue output ("In-IRef) feeds into following bit which performs exactly the same function. The process is repeated as many times as necessary in achieve the desired resolution.

6. Figures 6(a) and 6(b) show the basic design of two integrated circuit precision integrators.

(a) Derive an expression for the time constant of each integrator. Assume that the sampled-data integrator of Figure 6(b) is driven by non-overlapping clocks and that the switches are ideal. Give advantages and disadvantages of continous-time versus sampled data integrators

[8]

Fully differential tuneable integrators with time constant:

$$\tau = \frac{C}{2\beta\left(V_{C1} - V_{C2}\right)}$$

Thus integrator time-constant can be tuned by V_{C1} or V_{C2}. **Fully differential has advantages of cancelling offsets, and common-mode voltages such as noise** in addition to cancelling non-linearity as demonstrated in the resistor realisation.

Switched Capacitors

$$\frac{V_O}{V_{in}} = \frac{1}{j\omega C_2 R_{eq}} = \frac{f_C C_1}{j\omega C_2} = \frac{1}{jf/f_O} \quad \text{Where} \quad f_O = \left[\frac{1}{2\pi}\right]\frac{C_1}{C_2 . f_C}$$

Time constant $\tau = \frac{1}{f_C}\frac{C_2}{C_1}$

(b) With the addition of a single switched capacitor to Figure 6(b) show how the integrator can be made lossy?

[4]

Answers

Lossy Integrator

Essentially a low-pass filter. Parallel resistance placed in feedback path of op-amp

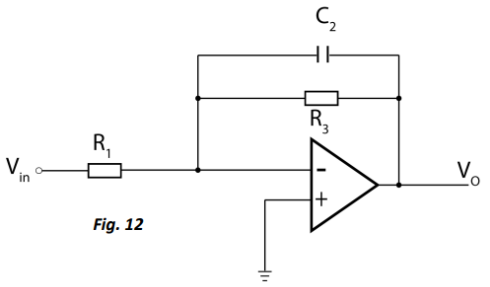


Fig. 12

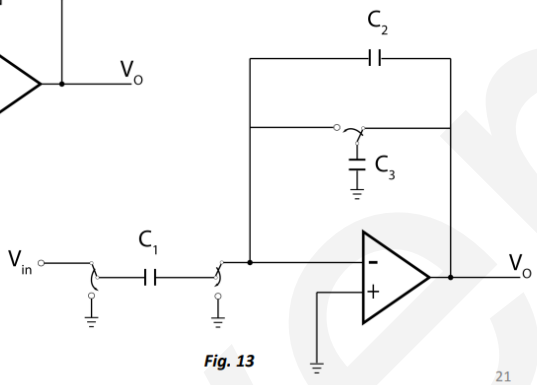


Fig. 13

21

(c) Show how the integrator of Figure 6(b) can be used to mimic both the inductor and capacitor of an LCR passive ladder filter.

[8]

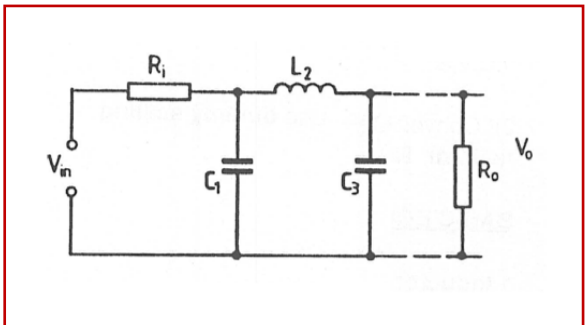
General transformation rules for ladder prototypes:

Inductor: $\frac{f_c L_2}{R_s} = \frac{C_{L2}}{C_u}$

Capacitor: $\frac{C_{c3}}{C_u} = f_c R_s C_3$

Resistor Rs=dummy scalar.

The circuit is equivalent to an RLC prototype



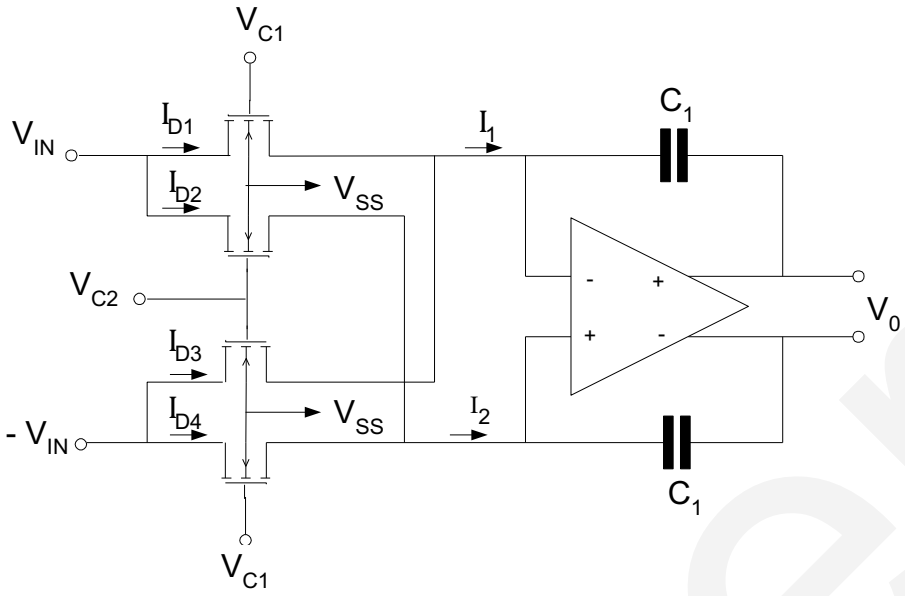


Figure 6(a)

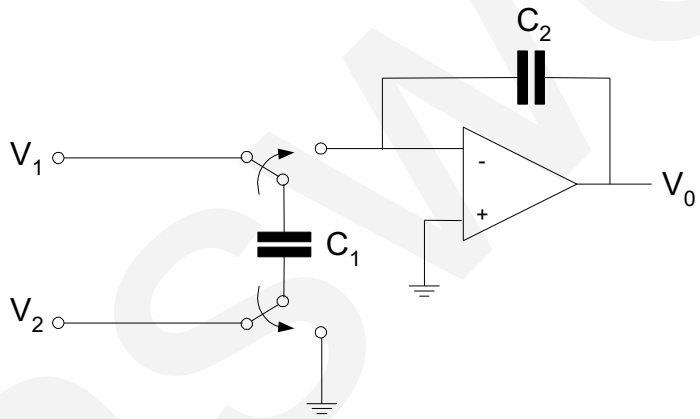


Figure 6(b)

Answers